

6.1.7. Let $A = \{x \in \mathbb{Z} \mid x = 6a + 4 \text{ for some integer } a\}$,
 $B = \{y \in \mathbb{Z} \mid y = 18b - 2 \text{ for some integer } b\}$, and
 $C = \{z \in \mathbb{Z} \mid z = 18c + 16 \text{ for some integer } c\}$.

Prove or disprove each of the following statements.

a. $A \subseteq B$

b. $B \subseteq A$

c. $B = C$

a. $A \subseteq B$

Proof:

Let $A = \{x \in \mathbb{Z} \mid x = 6a + 4\}$

Let $B = \{y \in \mathbb{Z} \mid y = 18b - 2\}$

Definition:

$$A \subseteq B \leftrightarrow \forall x, x \in A \rightarrow x \in B$$

Suppose $x \in A$. We must show that $\forall x \in A$,
if $x \in A$, then $x \in B$.

Let $a, b \in \mathbb{Z}$

$$x = 6a + 4$$

$$y = 18b - 2$$

$$x = y$$

$$6a + 4 = 18b - 2$$

$$6a = 18b - 6$$

$$a = 3b - 1$$

Let $a = 0$

$$6(0) + 4 = 4 \in \mathbb{Z}$$

$$3b - 1 = 0$$

$$3b = 1$$

$$b = 1/3 \notin \mathbb{Z}$$

Therefore, $A \not\subseteq B$

Solution: $A \not\subseteq B$

b. $B \subseteq A$

Let $B = \{y \in \mathbb{Z} \mid y = 18b - 2\}$

Let $A = \{x \in \mathbb{Z} \mid x = 6a + 4\}$

Definition:

$$B \subseteq A \leftrightarrow \forall y, y \in B \rightarrow y \in A$$

Suppose $y \in B$. We must show that $\forall y \in B$,

if $y \in B$, then $y \in A$.

Let $b, a \in \mathbb{Z}$

$$y = 18b - 2$$

$$x = 6a + 4$$

$$y = x$$

$$a = 3b - 1 \in \mathbb{Z}$$

Therefore, $B \subseteq A$

Solution: $B \subseteq A$

c. $B = C$

$$B = \{y \in \mathbb{Z} \mid y = 18b - 2\}$$

$$C = \{z \in \mathbb{Z} \mid z = 18c + 16\}$$

Suppose $B = C$. We must show that $B \subseteq C$
and $C \subseteq B$.

Let $b, a \in \mathbb{Z}$

$$y = 18b - 2$$

$$z = 18c + 16$$

$$y = z$$

$$18b - 2 = 18c + 16$$

$$18b = 18c + 18$$

$$b = c + 1 \in \mathbb{Z}$$

$$c = b - 1 \in \mathbb{Z}$$

Therefore, $B = C$.

Solution: $B = C$